Cambridge (CIE) IGCSE Physics
Paper 2: Multiple Choice (Set D)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/ s^2).

Scan here to mark your mock exam

or visit the mock exams landing page for this course

1 Identify the physical quantity that is **not** scalar.

A. speed

B. time

C. weight

D. energy

(1 mark)

2 A car travels along a clear 10.0 km section of motorway in 6.0 minutes. It then drives through 3.0 km of roadworks in 3.0 minutes.

Which calculation will give the correct average speed for the journey?

A. $\frac{3.0}{3.0}$ = 1.00 km/min

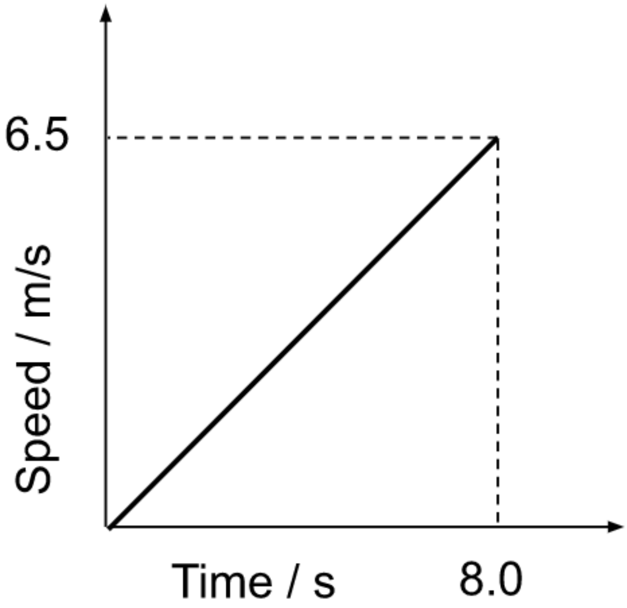
B. $\frac{10.0}{6.0}$ = 1.67 km/min

C. $1.67 + 1.00$ = 2.67 km/min

D. $\frac{13.0}{9.0}$ = 1.44 km/min

(1 mark)

3 The graph shows the journey undertaken by a car.



Which equation correctly gives the distance travelled by the car?

A. $\frac{6.5 \times 8.0}{2} = 26\text{m}$

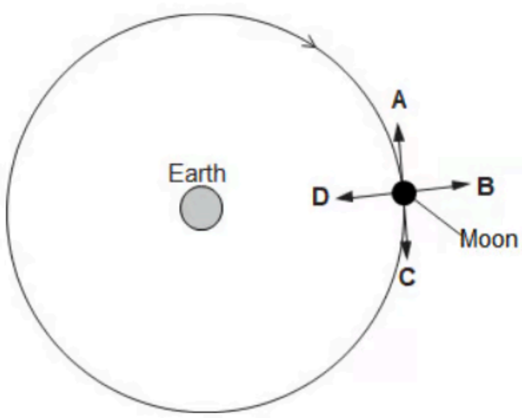
B. $6.5 \times 8.0 = 52 \text{ m}$

C. $\frac{6.5}{8.0} = 0.81 \text{ m}$

D. $\frac{8.0}{6.5} = 1.2 \text{ m}$

(1 mark)

4 The Moon orbits the Earth at a constant speed.



Which of the arrows on the diagram shows the direction of the resultant force on the Moon?

(1 mark)

5 A teacher has a hot cup of tea.

Some of the liquid in the cup evaporates.

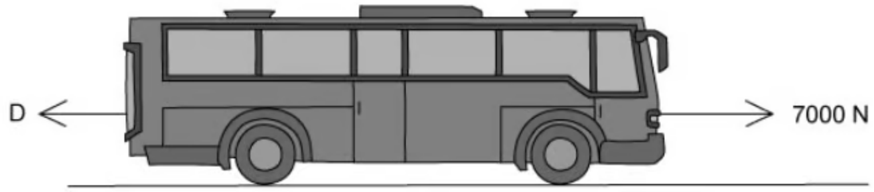
Choose the line from the table below that correctly describes what happens to the mass and the weight of the tea in the cup.

	Mass	**Weight**
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

(1 mark)

6 A bus drives along a flat road. Its engine produces a forward driving force of 7000 N. Drag forces, *D* also act on the bus.

The bus has an acceleration of 1.2 m/s^2 and a mass of 2500 kg.



What is the value of D?

A. 3000 N

B. 4000 N

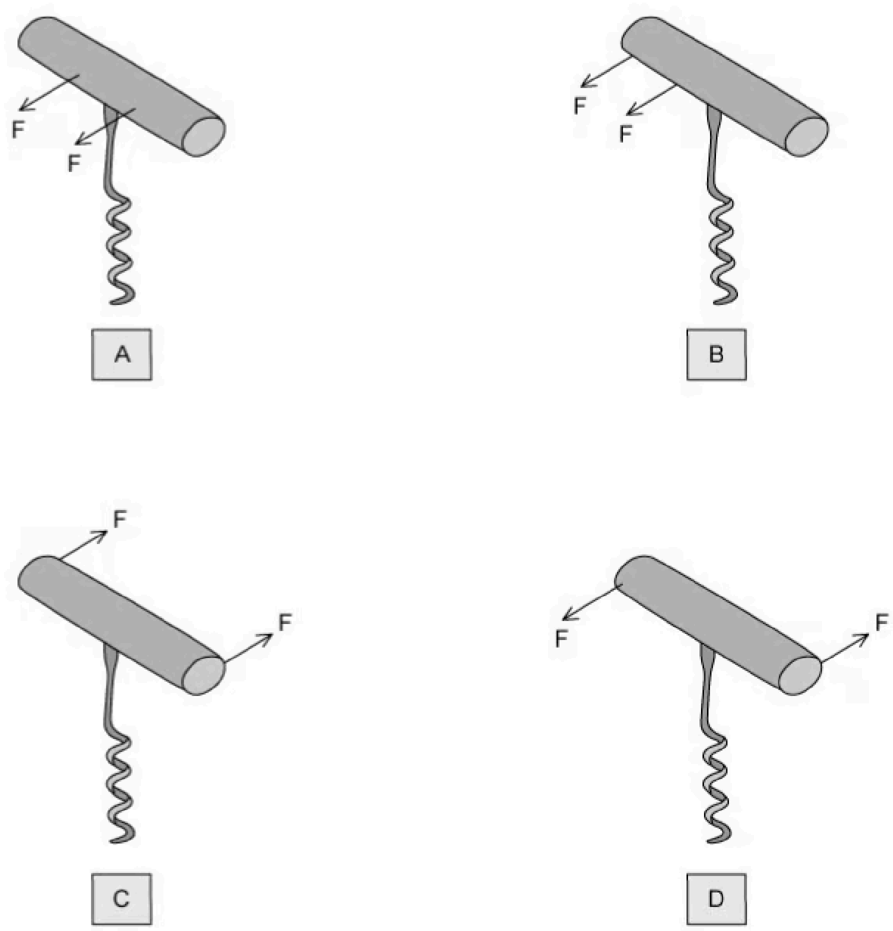
C. 4500 N

D. 5000 N

(1 mark)

7 A corkscrew is used to open a bottle of wine.

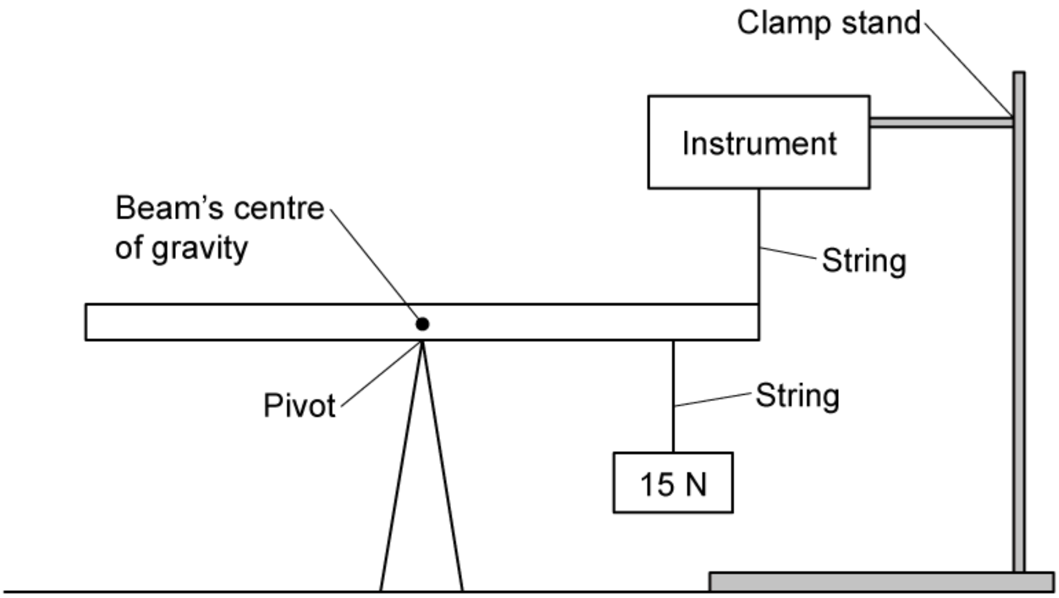
Two forces are applied at each of the positions shown below.



Which pair of forces produces the greatest turning force?

(1 mark)

8 In a classroom, a uniform beam with a pivot at its centre has a 15 N weight suspended from one end. An instrument fixed to the ceiling is attached to the same end.



What instrument must be used to show that the resultant moment on a system in equilibrium is zero?

A. Joulemeter

B. Force meter

C. Ruler

D. Momentmeter

(1 mark)

9 A bullet is shot from a gun. The bullet moves forward and the gun moves in the opposite direction. Which of the following statements is true?

A. They move with the same velocity in opposite directions.

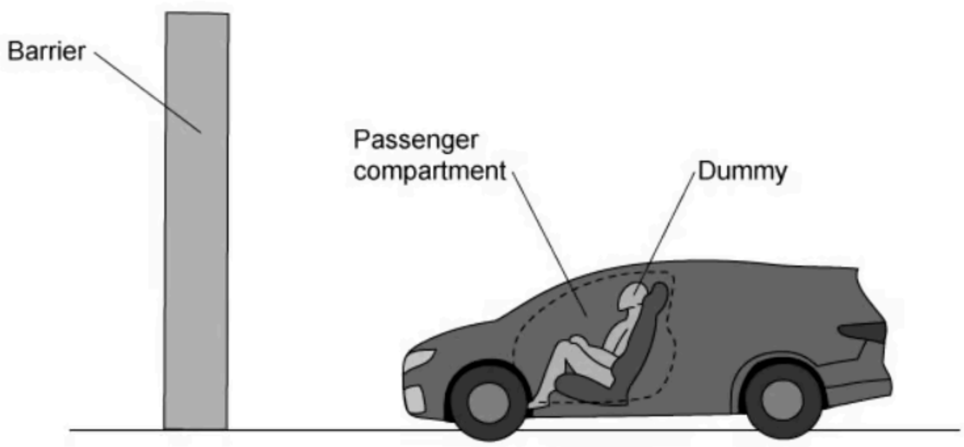
B. The bullet moves with a slower velocity due to its mass.

C. The total momentum of the system is zero.

D. The total momentum of the system does not stay constant before and after the collision.

(1 mark)

10 A dummy is used in a crash test to investigate the safety of a new car. The deceleration of the dummy and the passenger compartment is measured.



The deceleration of the dummy is less than the deceleration of the passenger compartment.

Why is this of benefit for the safety of the passenger?

A. Seat belts, crumple zones and airbags all absorb energy from the crash

B. The crash happens quickly so the passenger doesn't feel anything

C. The force of impact on the passenger, and therefore harm, is reduced

D. The time over which the collision happens is elongated for the passenger

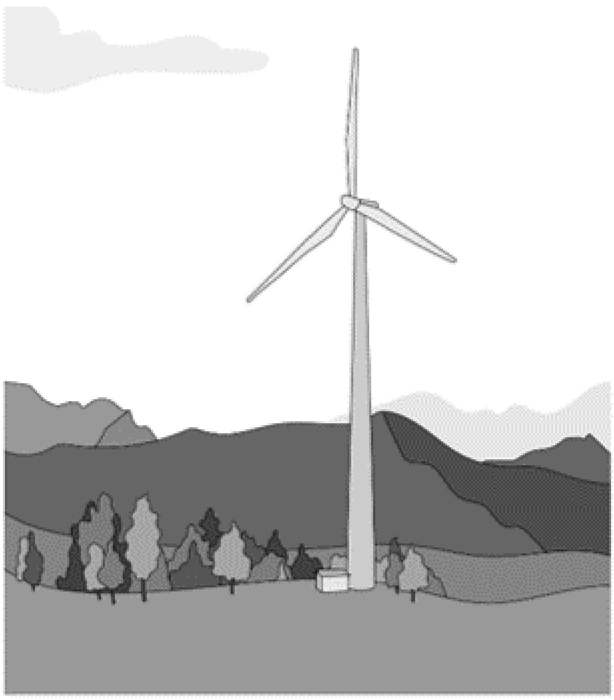
(1 mark)

11 Which row correctly matches an energy store with an example of that store?

	Energy Store	Example
A.	kinetic	a car moving at 70 km / h
B.	elastic	a box sitting high on a shelf
C.	magnetic	compression of springs in a car's suspension
D.	chemical	the nuclei of uranium atoms

(1 mark)

12 The diagram below shows a wind turbine.



What is the change in the kinetic energy if the wind speed is reduced by half?

A. $E_K = \frac{1}{8}$

B. $E_K = \frac{1}{4}$

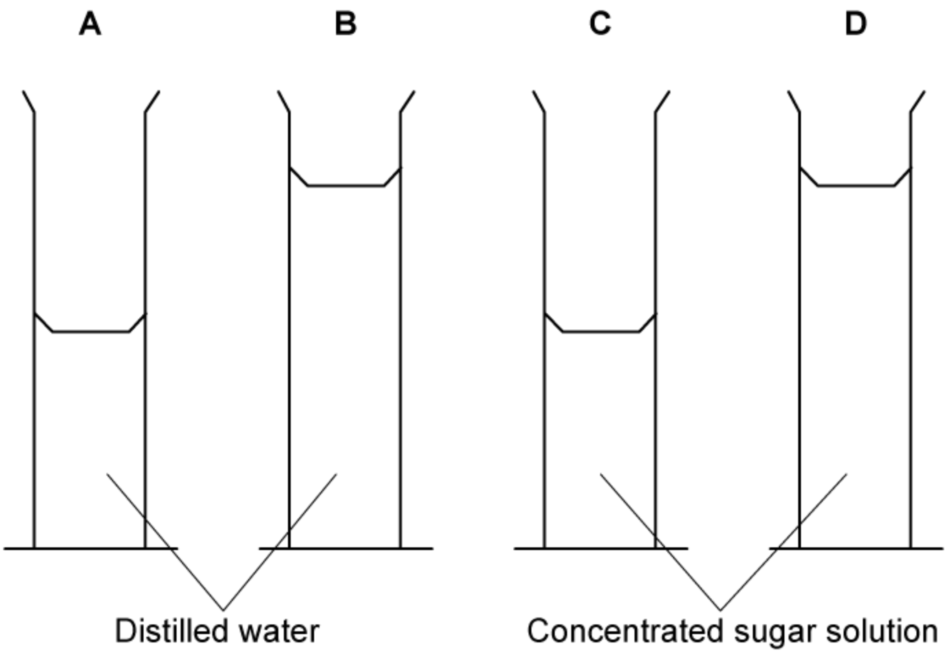
C. $E_K = \frac{1}{2}$

D. No change

(1 mark)

13 The diagram shows four measuring identical cylinders containing either distilled water or concentrated sugar solution.

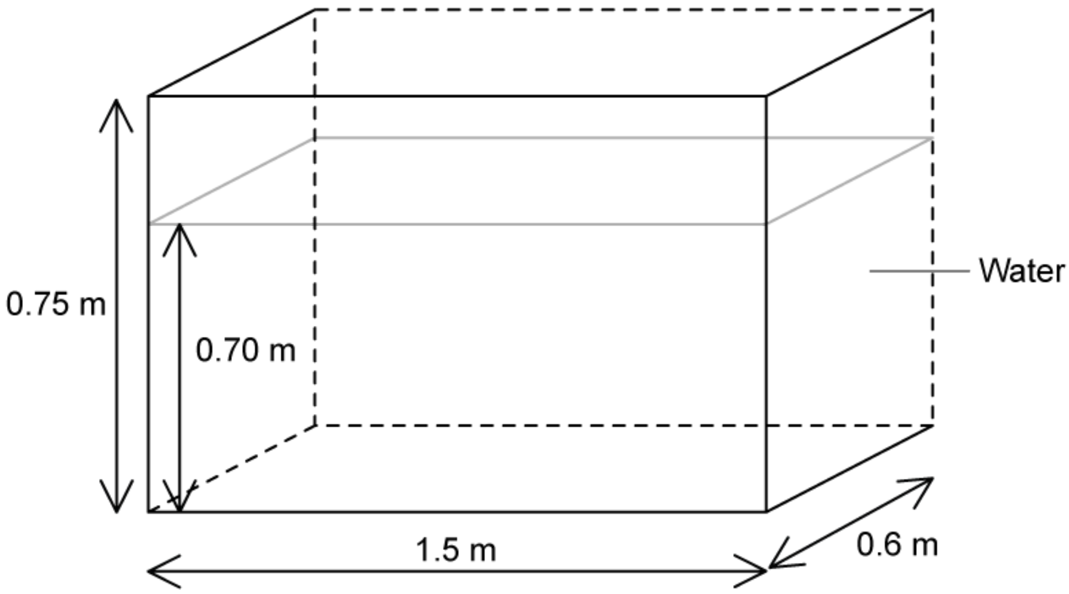
Which measuring cylinder has the least pressure at the base, due to the liquid?



(1 mark)

14 For the tank of water in the diagram below, which value gives the pressure on the base of the tank due to the water?

The density of water = 1000 kg/m^3



A. 6500 Pa

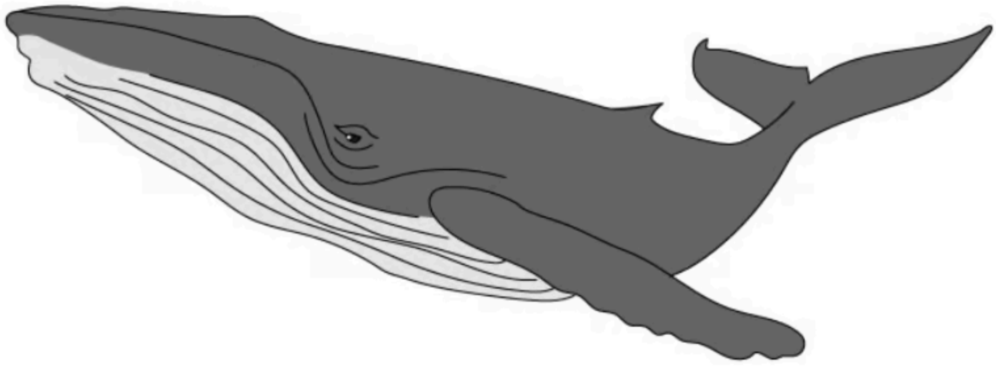
B. 6900 Pa

C. 7400 Pa

D. 7900 Pa

(1 mark)

15 A whale takes a breath at the surface of the ocean, where the pressure is 1.00×10^5 Pa. It then dives to a depth where the pressure is 1.21×10^5 Pa. The temperature is the same at this depth as it is at the surface.



The whale lets out a bubble of air of volume 50 cm^3. This bubble rises to the surface of the ocean.

What is the volume of the bubble when it reaches the surface?

A. 41.3 cm^3

B. 60.5 cm^3

C. 50.0 cm^3

D. 21.0 cm^3

(1 mark)

16 A pan, containing 0.750 kg of water which is initially at 13 ℃ is heated by an electric hob. 35 kJ of thermal energy is put into the water and its temperature rises.

You can assume that all the energy supplied by the hob goes into raising the temperature of the water.

The specific heat capacity of water is 4200 J/kg ℃.

To the nearest ℃, what is the final temperature of the water?

A. 24 ℃

B. 13 ℃

C. 19 ℃

D. 11 ℃

(1 mark)

17 Internal energy is affected by two properties of the particles which make up a substance.

Identify the two properties.

A. Motion and position

B. Acceleration and mass

C. Specific heat capacity and volume

D. State of matter and potential energy

(1 mark)

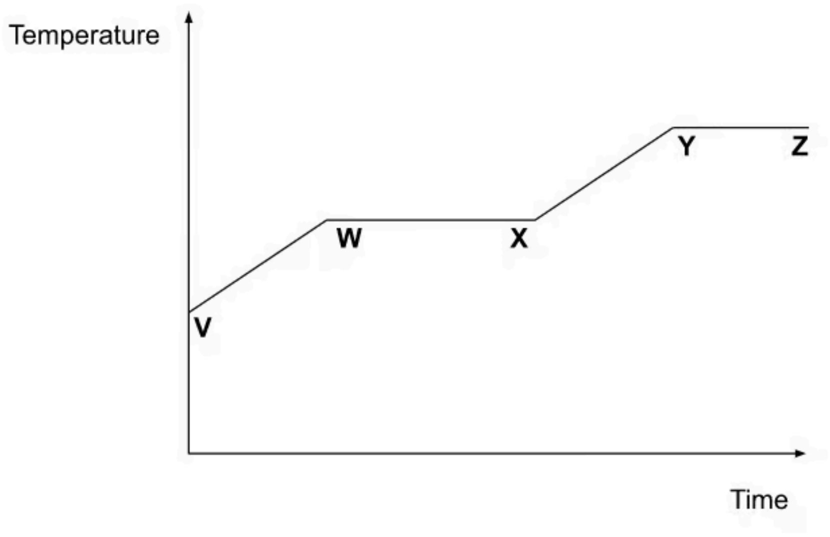
18 Which of the following processes take place in a non-metal to allow the **conduction** of thermal energy?

	Transfer of electrons	**Lattice vibrations**	**Transfer of nuclei**
A	yes	yes	no
B	no	yes	no
C	yes	yes	yes
D	no	yes	yes

(1 mark)

19 A substance, which is initially in its solid form, is heated at a constant rate.

The graph shows how its temperature changes with time.



Between which points is the substance partly a liquid and partly a gas?

A. V and W

B. W and X

C. X and Y

D. Y and Z

(1 mark)

20 Which of the following statements about convection is correct?

Warm air rises because:

A. The air molecules expand and become less dense.

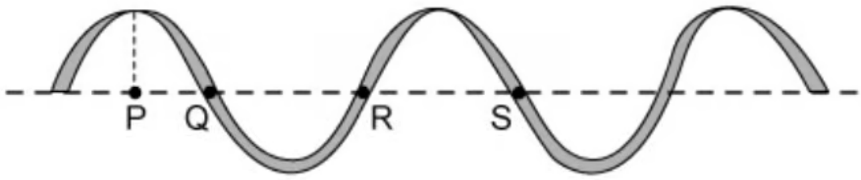
B. The air molecules move further apart, making the air less dense.

C. The air molecules contract, making the air less dense.

D. The air molecules move closer together, making the air less dense.

(1 mark)

21 Which two points on the wave are exactly one wavelength apart?



A. P and S

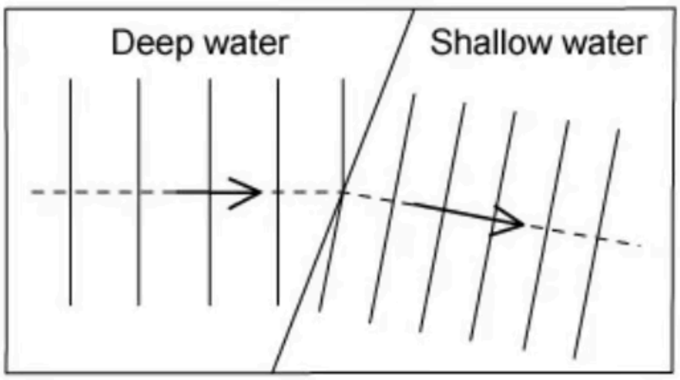
B. Q and R

C. Q and S

D. P and R

(1 mark)

22 When water waves travel from deeper water into more shallow water they are refracted.



Why does the wave change direction in the manner shown?

A. The wave speed increases.

B. The wave speed decreases.

C. The frequency increases.

D. The frequency decreases.

(1 mark)

23 When light passes from a denser material into a less dense material

A. light slows down and bends toward the normal

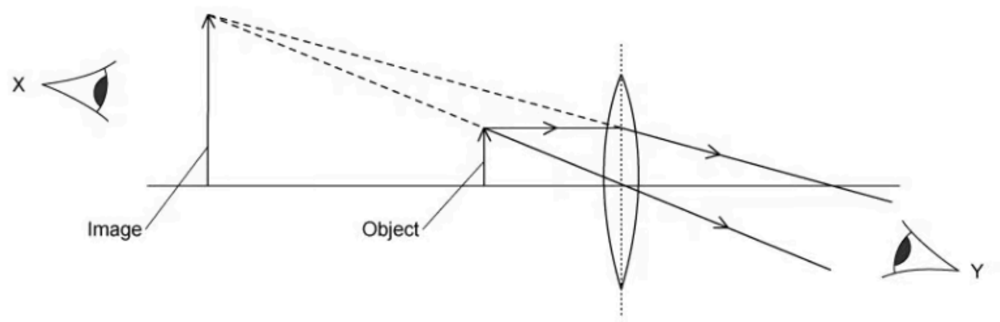
B. light slows down and bends away from the normal

C. light speeds up and bends toward the normal

D. light speeds up and bends away from the normal

(1 mark)

24 Below is a ray diagram showing the image formed by a thin, converging lens.



Which of the following statements about the image is correct?

A. The image is virtual and can be seen by eye Y.

B. The image is inverted and can be seen by eye Y.

C. The image is real and can be seen by the eye at X.

D. The image is virtual and can be seen by eye X.

(1 mark)

25 Light enters a flat water surface at an angle of 27 °.

The refractive index of water is 1.33.

At what angle does the light ray refract?

A. 0.34 °

B. 20.3 °

C. 20.0 °

D. 37.0 °

(1 mark)

26 The table below shows the entire electromagnetic spectrum. It runs in order of increasing wavelength.

Which region is microwaves?

γ-rays	**A**	**B**	**C**	infrared	**D**	radio waves

(1 mark)

27 Which type of electromagnetic wave is used in remote controls?

A. Microwaves

B. Visible light

C. Ultraviolet

D. Infrared

(1 mark)

28 A student makes a series of statements about a permanent bar magnet.

Which of them is true?

A. The North pole of a permanent bar magnet will repel the North pole of another magnet.

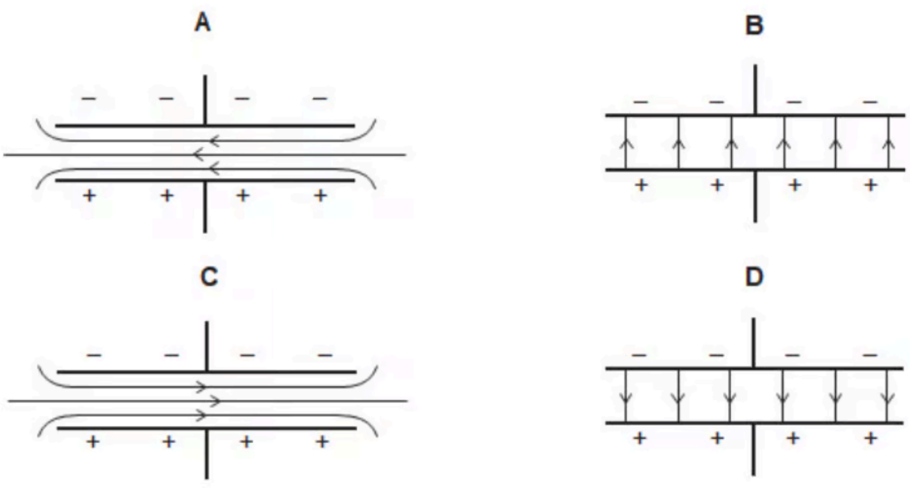
B. A permanent bar magnet must be made from a soft magnetic material.

C. A permanent bar magnet can repel a magnetic material.

D. The field lines of a permanent bar magnet cross each other where the field is strongest.

(1 mark)

29 Which diagram shows the correct electric field pattern between two oppositely charged parallel plates?



(1 mark)

30 What is the correct unit of charge?

A. Ampere

B. Coulomb

C. Newton

D. Volt

(1 mark)

31 A student considers how changing the diameter and length of a wire affects its resistance.

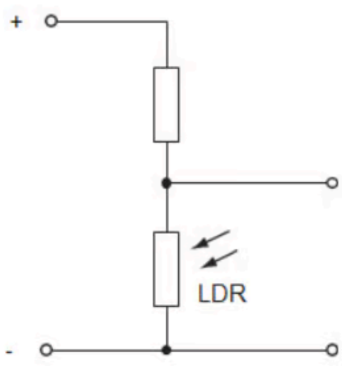
Which row shows the changes that could be made to the length and diameter of the wire to **decrease** its resistance?

	Change to length	**Change to diameter**
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

(1 mark)

32 A homeowner buys a security light that switches on automatically when it gets dark.

The security light contains the circuit, shown below.

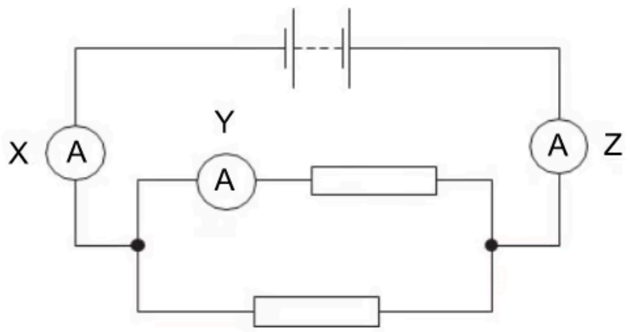


When it gets dark, what happens to the resistance of the LDR, and what happens to the potential difference across the LDR?

	Resistance of LDR	p.d. across LDR
A	increases	increases
B	decreases	increases
C	increases	decreases
D	decreases	decreases

(1 mark)

33 Which of the following statements about the current in ammeters **X**, **Y** and **Z** is correct?



A. The current through X equals the current through Y + the current through Z.

B. The current through X is equal to the current through Z.

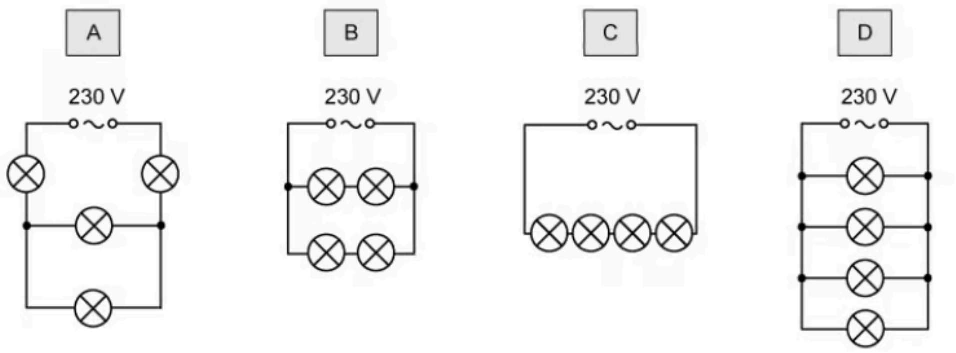
C. The current through X is equal to the current through Y.

D. The current through Y is greater than the current through Z.

(1 mark)

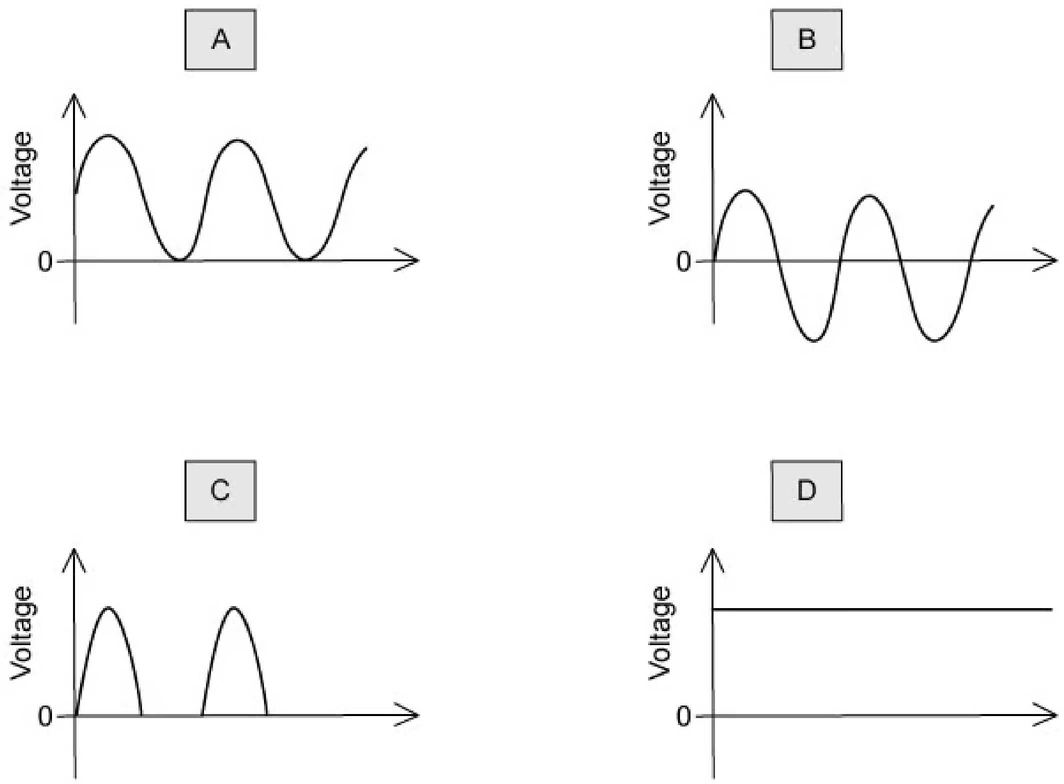
34 All the light bulbs in the diagrams above are identical. Each one is rated as ‘10 W, 230 V’.

In which of the diagrams are all the bulbs working at their normal brightness?



(1 mark)

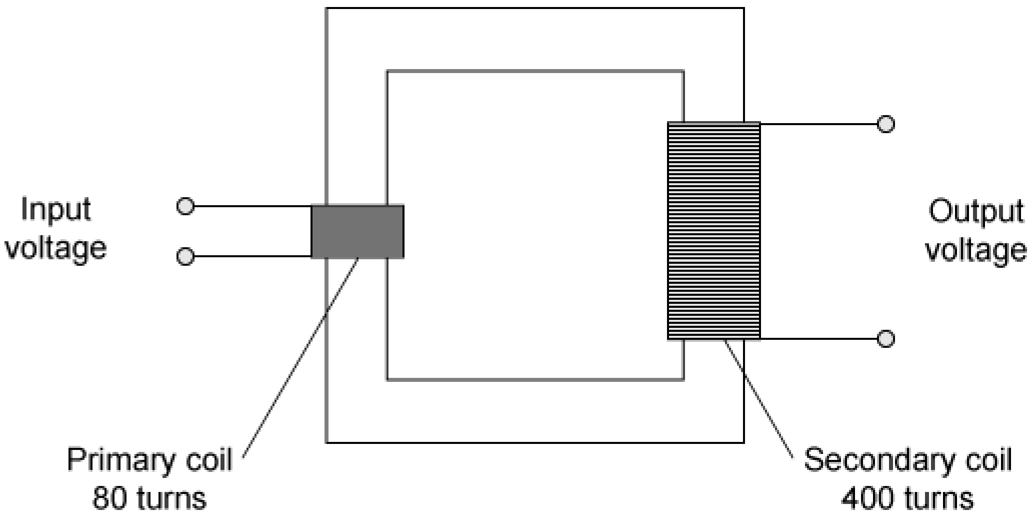
35 Which graph represents the output of a simple a.c. generator?



(1 mark)

36 Transformers are used to change the voltage of a power supply.

The diagram below shows a simple transformer.



If the input voltage is 48 V, what is the output voltage?

A. 2.4 V

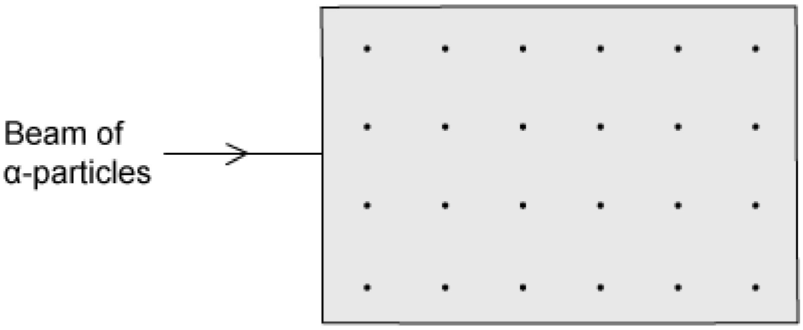
B. 9.6 V

C. 240 V

D. 667 V

(1 mark)

37 A magnetic field can be represented by the diagram shown below. The dots represent magnetic field lines coming out of the page.



A beam of alpha particles is directed through the field as shown above. Alpha particles, being charged, will be deflected by the field.

In which direction will the alpha particles be deflected?

A. upwards

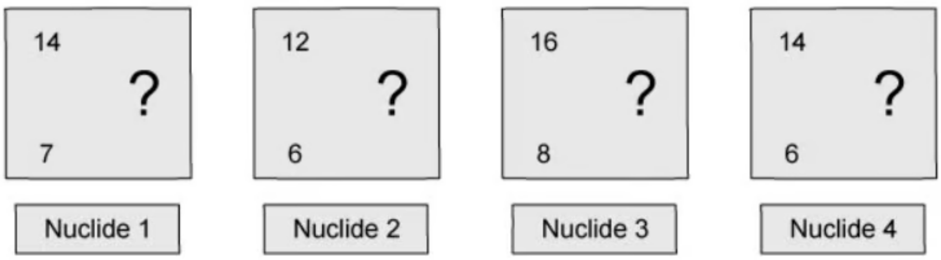
B. downwards

C. into the page

D. out of the page

(1 mark)

38 Below are the symbols for four different nuclides.



Which two nuclides are isotopes of the same element?

A. Nuclide 1 and nuclide 4.

B. Nuclide 1, nuclide 3 and nuclide 4

C. Nuclide 2 and nuclide 3

D. Nuclide 2 and nuclide 4

(1 mark)

39 Which statement about **α**-particles is correct?

A. **α-**particles consist of two protons and two electrons.

B. **α-**particles are the most penetrating type of radiation.

C. **α-**particles are a type of electromagnetic radiation.

D. **α**-particles are highly ionising.

(1 mark)

40 Strontium-90 is a radioactive substance with the nuclide symbol $^{90}_{38}Sr$.

It decays by emitting radiation, as shown by the following equation.

$$^{90}_{38}Sr \rightarrow ^{90}_{39}Y + ?$$

What is missing in this equation?

A. **α-**particle

B. Neutron

C. **γ**-ray

D. **β**-particle

(1 mark)